

Authors

Jason Yang, Rocco Vitela, Annika Terezakis, Panashe Nhenga, Audrey Hubbell

Sub-Problem

- Each female Spotted Lantern Fly (SLF) lays ~3 egg masses of 30–60 eggs each
- Rapid reproduction enables popoulation spikes within a single season
- SLFs feed on sap, which weakens vines and could kill plants in severe cases
- Removing egg masses limits reproduction and lowers seasonal infestations

Solution

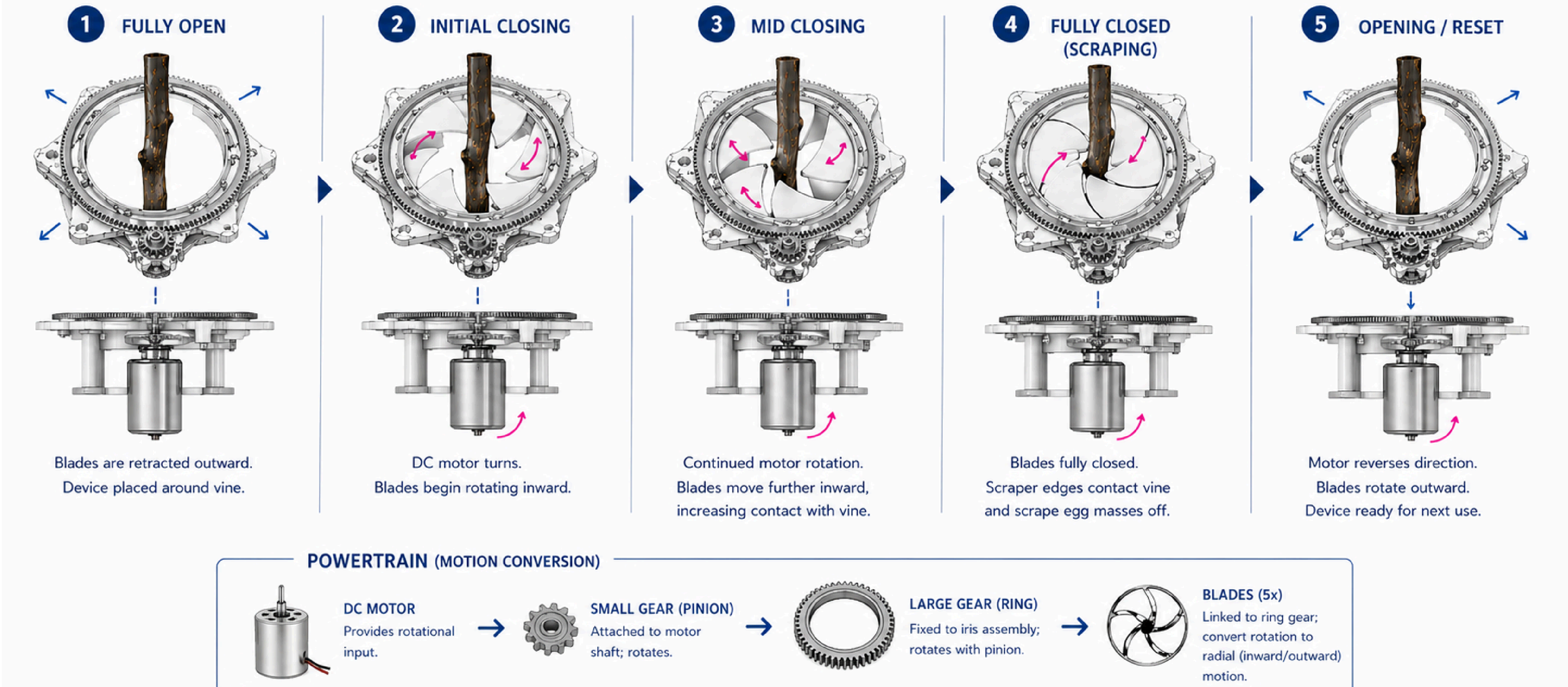
- Handheld radial compression device for vine and branch use
- Slotted blades to effectively remove egg masses
- DC motor coupled with a gear system to open and close the device for inward radial compression
- Applies controlled force to crush and destroy eggs while minimizing plant damage
- Plant-safe and scalable for use across vineyards and similar environments

Final Prototype

- 3D printed PLA structure which include the Base, Standoffs, Gears, and Blades
- Uses a DC motor connected to a gear pinion
- Motor driven system enables opening and closing the mechanism through use of a switch
- Blade design optimized to encase radial objects such as grape vines and tree branches
- Designed to effectively scrape and remove egg masses

IRIS MECHANISM MOTION

Rotational input from the DC motor (vertically mounted) is converted to radial motion of five blades, enabling the device to open, close, and scrape egg masses from grape vines.



Testing Methodology

1 Force Required Test

The mechanism required ~5–7 N of force to open & 10–12 N to fully close. These force levels fall well within the range of what a user can comfortably apply by hand. For comparison, everyday actions such as pressing a button or pulling a small drawer typically require a force of 5–15 N.

2 Egg Mass Removal Effectiveness

The radial compression mechanism was able to successfully apply force to the simulated egg masses and effectively scrape them from the vine model in 100% of our trials (3/3).

3 Repeated Use

After 20 trials, we noticed no issues with the radial motion piece. After ~10 trials, the string attached to the pulley loosened, and eventually popped clean off, which led us to convert our mechanism operation to be motor-controlled. The range of applied force varied from 5–10N throughout trials.

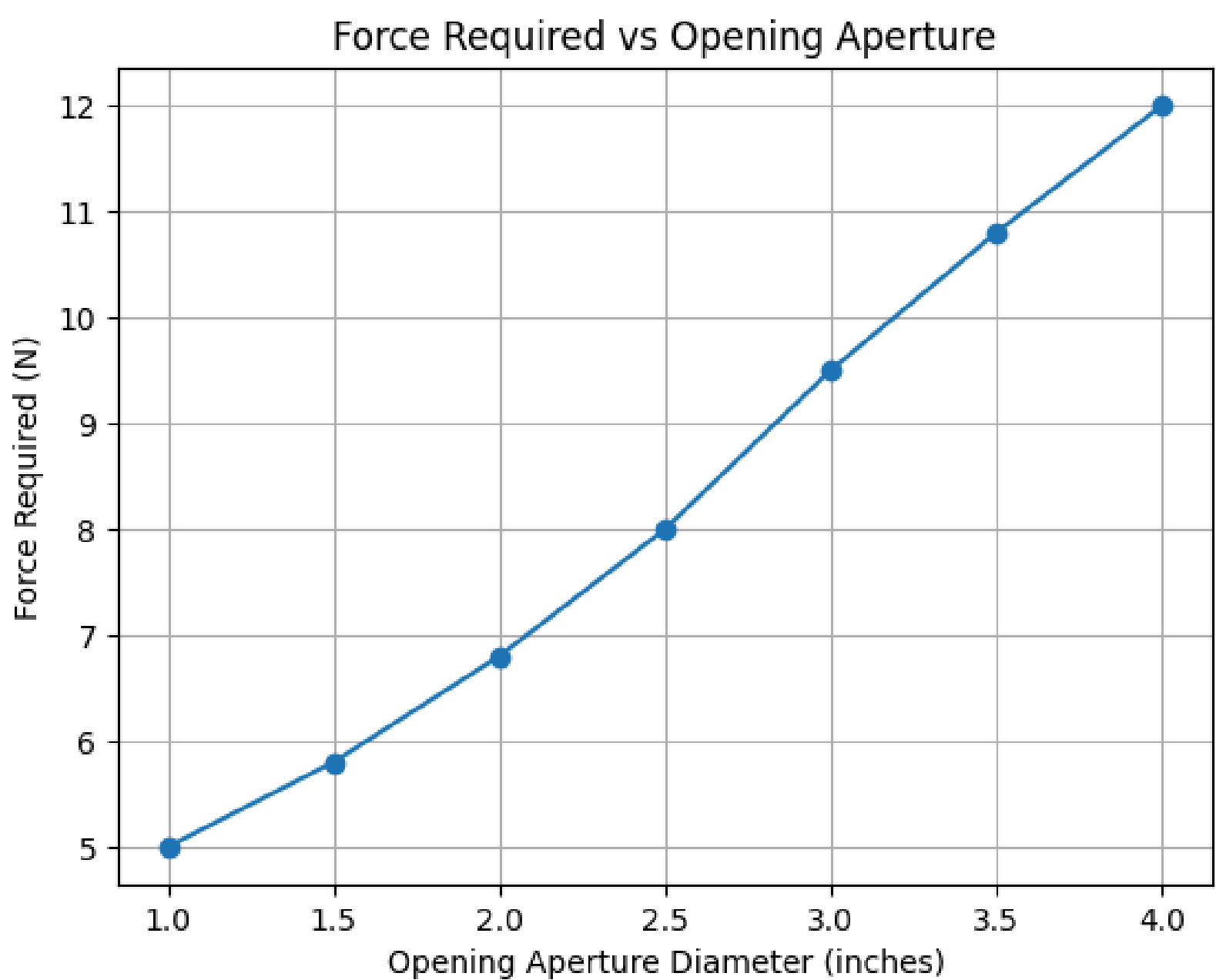
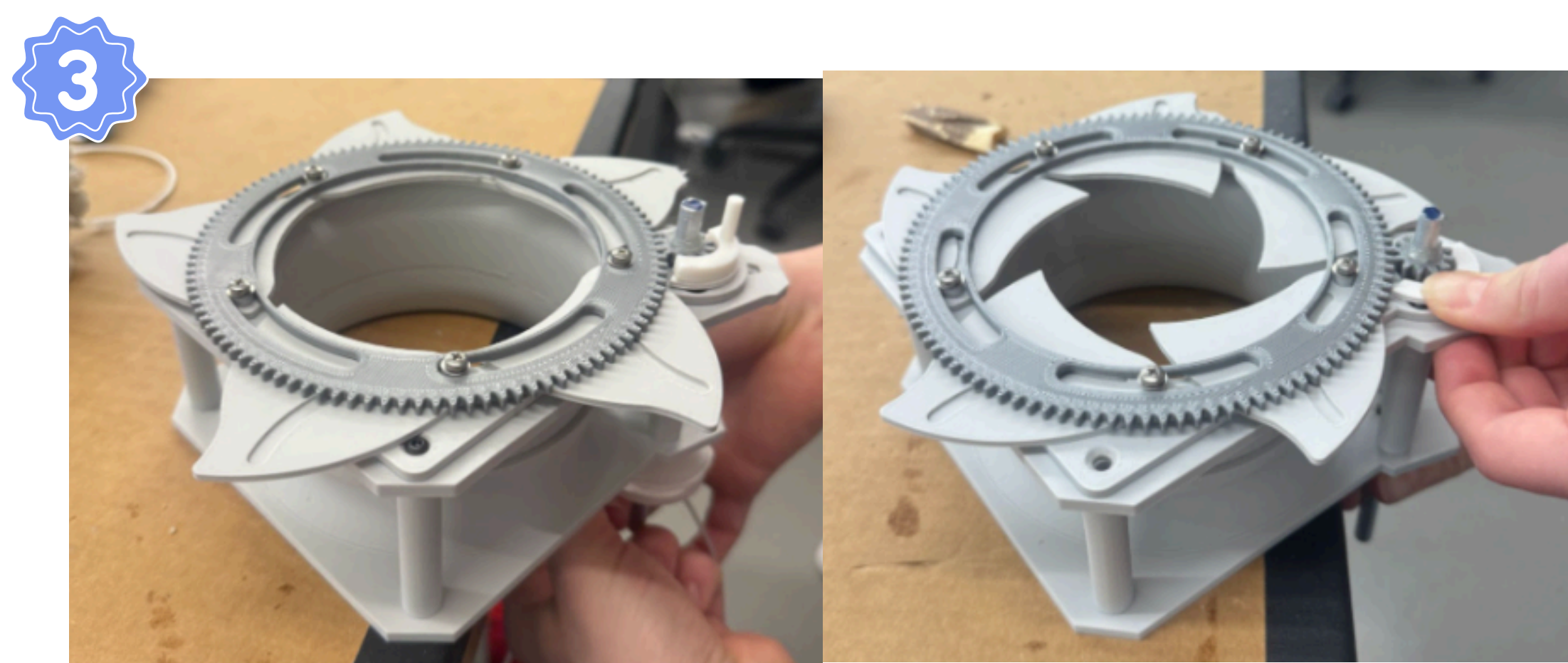


Figure 1: Force Required to Open Iris Mechanism vs. Open Aperture Diameter

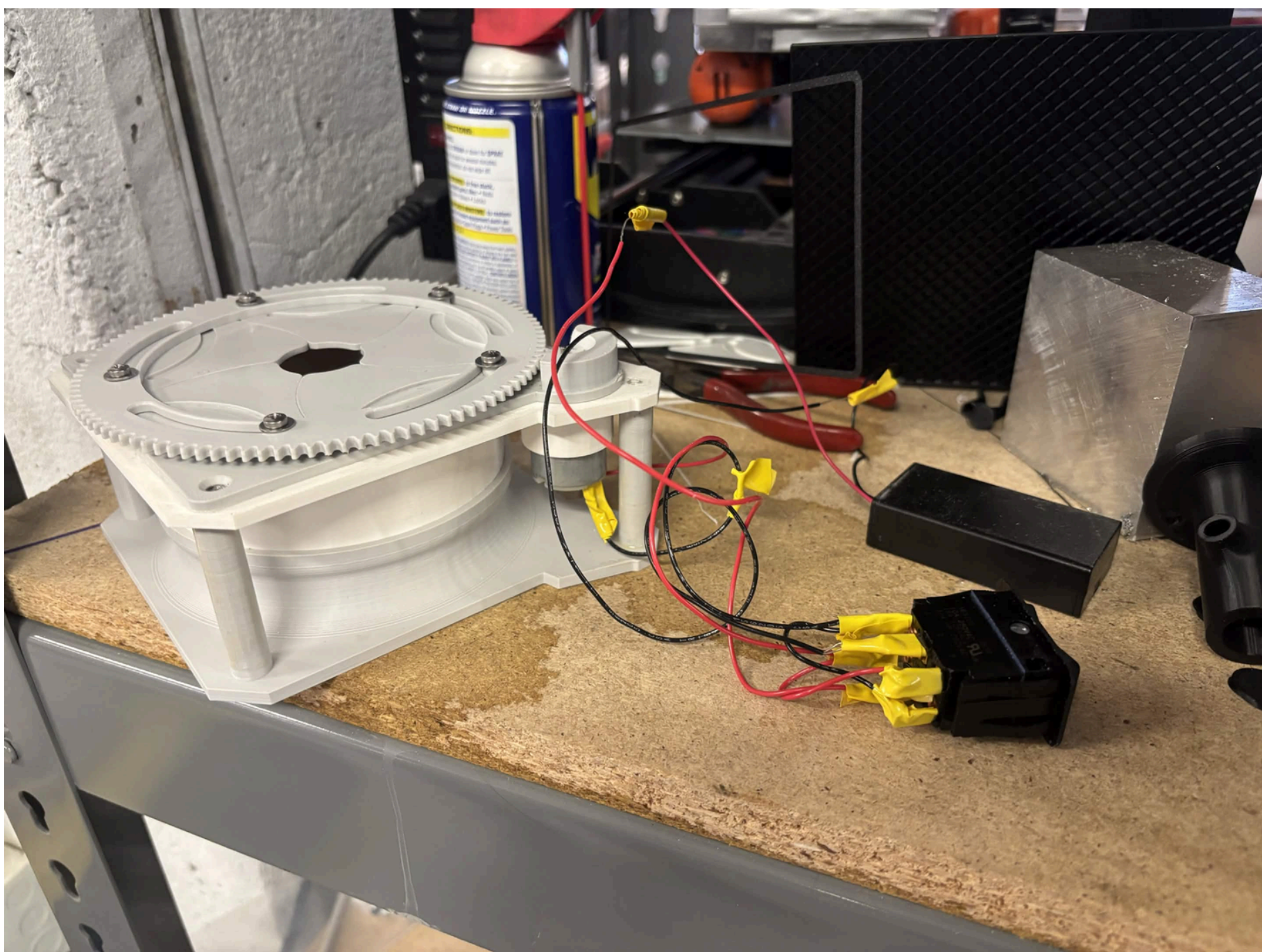


Future Steps

- Hinge mechanism
- Telescoping Rod
- Sturdier Aluminum Construction
- Stronger Torque Motor
- Add Compliant Inner-Blade Material

Conclusions

In conclusion, we have found that our prototype works effectively as a handheld tool to help radially scrape off egg masses on various contact surfaces. We have developed a method to electrically open and close our blade apparatus, so that the primary effort provided by a human is the physical scraping of the vines. We have performed various tests, all of which indicate that our structure is physically stable enough to perform these tasks and that repeated repairs are feasible to undertake.



Key Sources & Acknowledgements

Ohio Department of Agriculture. (n.d.). Spotted lanternfly best management practices. <http://agri.ohio.gov/divisions/plant-health/invasive-pests/slf-bmp>
Penn State Extension. (n.d.). What should you do with spotted lanternfly egg masses?
List the primary sources used during the research, such as academic articles, books, and websites.

Thank you to the MAE 2250 Course Staff, Cornell CALS Extension, E&J Gallo Winery, and National Grape for their support throughout this project

SORRY LITTLE FLY

T3

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MAE 2250

Sub-Problem

Spotted lanternflies reproduce rapidly; each female lays 3 egg masses of 30–60 eggs, allowing populations to explode in a single season. SLFs feed on sap, weakening vines, reducing cold hardiness, and potentially killing plants. Destroying egg masses reduces SLF reproduction, lowering seasonal infestations and protecting yield and vine health.

Solution

A handheld, radial compression device that fits around the vine, with slotted blades to remove egg masses. A DC motor opens, then closes the device through gears for uniform inward radial compression, crushing and destroying eggs without damaging the plant. A plant-safe & scalable tool, our device could be used across vineyards and other sites.

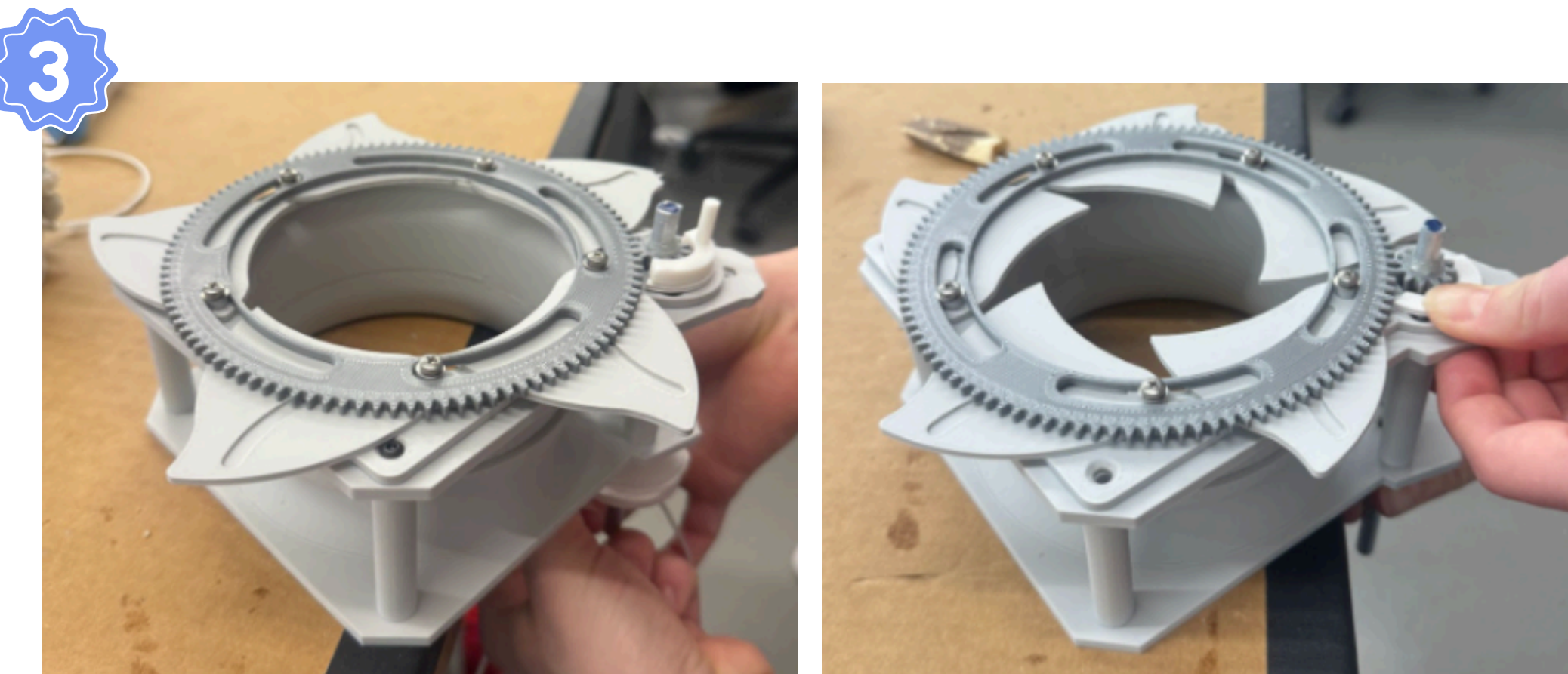
Testing Methodology

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- Egg Mass Removal Effectiveness**

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Final Prototype

Our final prototype consists of a 3D-printed body structure made from PLA, including the base, standoffs, gears, and blades of the system. We use a driving shaft from McMaster, geared to a DC motor, to open and close the mechanism. The blade designs have been finalized to better encase radial objects, such as grape vines and tree branches, from which we are scraping eggs.

Conclusions

In conclusion, we have found that our prototype works effectively as a handheld tool to help radially scrape off egg masses on various contact surfaces. We have developed a method to electrically open and close our blade apparatus, so that the primary effort provided by a human is the physical scraping of the vines. We have performed various tests, all of which indicate that our structure is physically stable enough to perform these tasks and that repeated repairs are feasible to undertake.

Future Steps

For future steps, we hope to incorporate a hinge mechanism that would allow the enclosure to fully open and encase branches of varying sizes. In addition, the development of a handle to better utilize this scraping mechanism in hard-to-reach areas would be a critical next step. Given sufficient time and resources, remaking the mechanism out of aluminum would also be a valid next step.



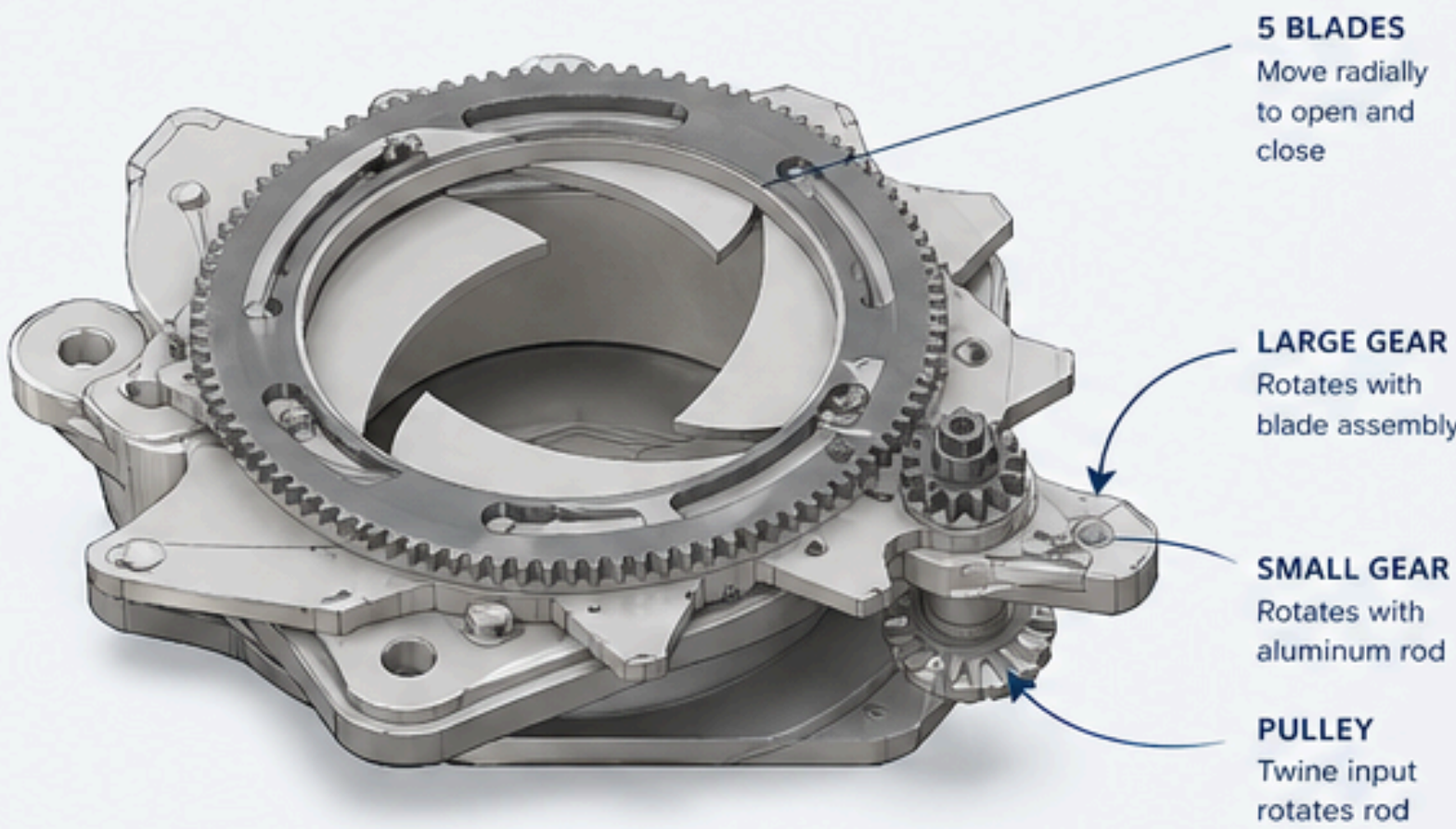
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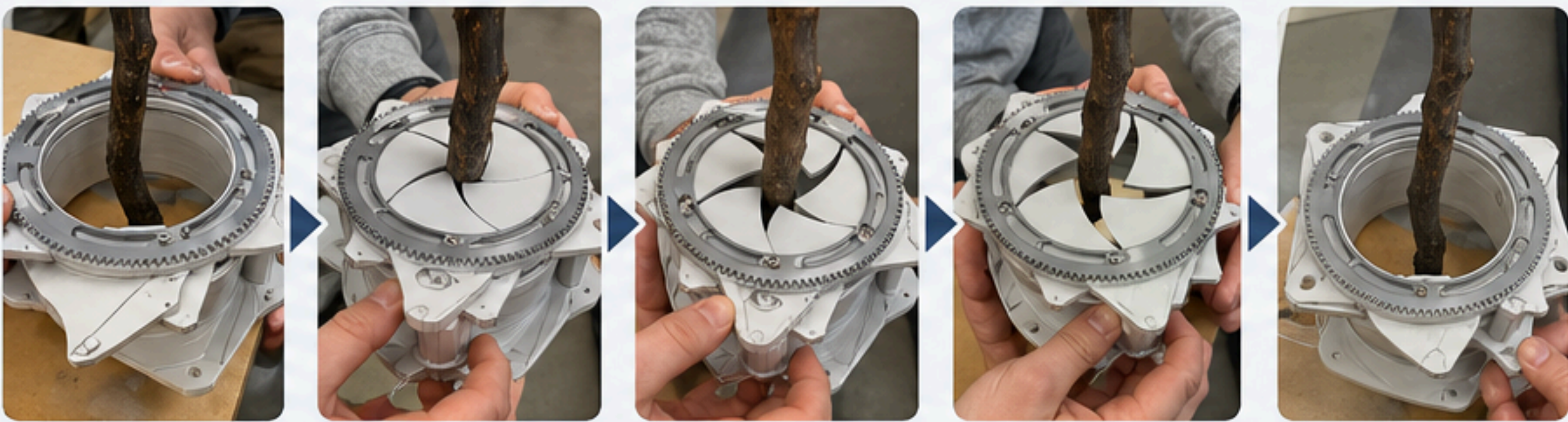
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IRIS MECHANISM MOTION

Rotational input from the pulley is converted to radial motion of five blades, enabling controlled opening and closing around grape vines to scrape egg masses.

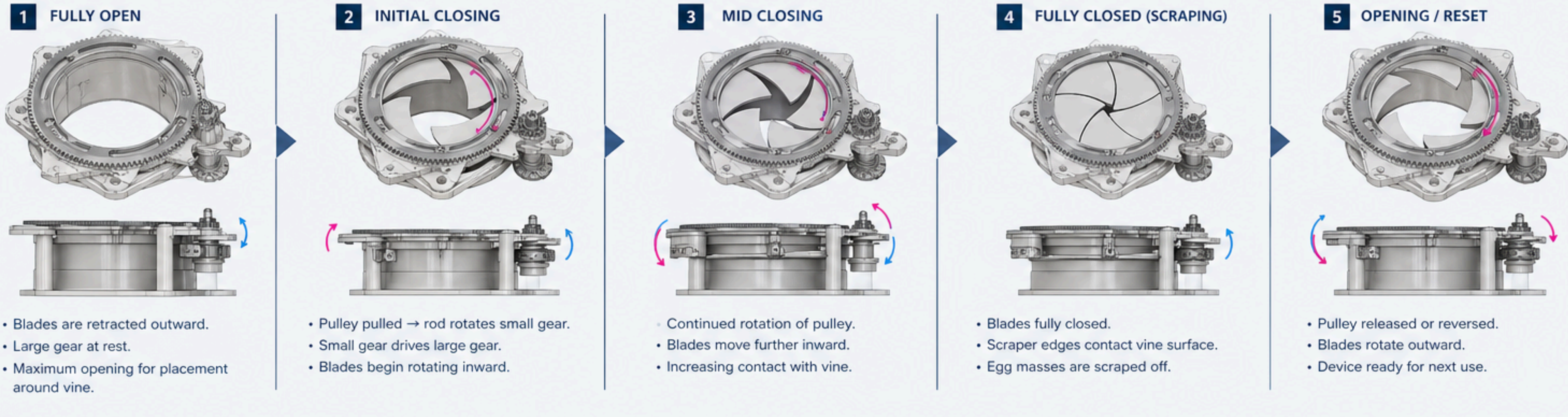


REAL-WORLD OPERATION



- 1. OPEN**
Blades retracted. Device placed around vine.
- 2. CLOSE**
Pulley is pulled, blades rotate inward to contact vine.
- 3. SCRAPE**
Continued closing scrapes egg masses from vine.
- 4. OPEN**
Pulley released, blades rotate outward and retract.
- 5. RELEASE**
Device removed from vine. Ready for next use.

MECHANISM MOTION: STEP-BY-STEP



KINEMATICS SUMMARY



IRIS MECHANISM MOTION

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